

Symmetrical Cognition Between Physical Humans and Virtual Agents

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ABSTRACT

Cognition is a crucial problem that affects human performance and subjective experiences in the given environment. However, the cognition problem will be different if we consider it in a symmetrical reality environment. Though researchers have studied the cognition problem for many years, there are still many different points within a symmetrical reality environment. In this paper, we discuss the cognition problem from some perspectives, *i.e.*, attention, perception, pattern recognition, and communication. We show how the symmetrical reality system can change the traditional attention mechanism, the perception process, the recognition and the communication between different agents. Finally, we introduce the application of symmetrical cognition to expand the current cognition research and give some useful suggestions for studying the cognition problem.

Index Terms: Computing methodologies—Artificial intelligence—Philosophical/theoretical foundations of artificial intelligence—Cognitive science; Human-centered computing—Human computer interaction (HCI)—HCI theory, concepts and models

1 PERCEPTUAL STRUCTURE OF SYMMETRICAL REALITY

Symmetrical reality [7] is the unified description of virtual reality, augmented reality, inverse virtual reality [6] and inverse augmented reality [8]. With the virtual agents based on independent minds, the symmetrical reality brings many new changes to the traditional research areas. In this paper, we will pay attention to cognition, which is a discipline to study how people can obtain knowledge from the physical world.

Since the so-called symmetrical reality world is different from the purely physical world, we should rethink cognition due to the new features of symmetrical reality. The most obvious feature is that there are two perceptual centers within the framework of symmetrical reality. One center is the physical human, who can perceive both the physical world and the virtual world. The other center is the virtual agent, who can perceive both the physical world and the virtual world as well. Because of the double-center perceptual structure, cognition is no longer what it used to be in the past years. Both physical humans and virtual agents could have cognitive abilities.

The research of cognition regarding physical humans has a complete architecture, so the cognition of virtual agents can also be studied using similar methods. When we refer to the cognition problems in symmetrical reality, we are always talking about the cognition of physical humans and the cognition of virtual agents. Next, we will discuss these problems of symmetrical cognition regarding attention, perception, recognition, and communication. Fig. 1 has shown the basic contents of symmetrical cognition.

2 COGNITION IN SYMMETRICAL REALITY

Cognition is a complicated process that refers to the acquisition, processing, storage, and application of information. Humans live in the physical world and use the cognitive ability to obtain knowledge from the surrounding environment. We highlight the differences of cognition within symmetrical reality to show how humans and

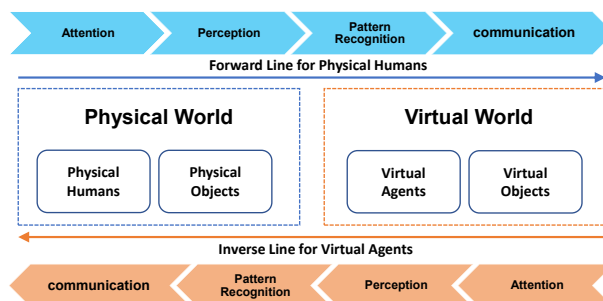


Figure 1: Symmetrical cognition based on the framework of symmetrical reality.

virtual agents deal with cognitive issues. We explore the contents in four aspects: 1) Attention; 2) Perception; 3) Pattern recognition; 4) Communication.

2.1 Attention

Attention [1] denotes the interesting area in the human's observation area. The cognition begins with human attention that is paid to a certain object. Usually, attention means where humans are observing, and this is the beginning of a cognitive process.

2.1.1 Effect of Agent

The virtual agent located in the virtual world has the ability to control the objects in the physical world, and it also has the physical embodiment in the physical world. For this reason, the physical world becomes different from what the human has ever been. So the attention mechanism in symmetrical reality is different from that in the purely physical world. In a symmetrical reality system, we need to make it clear that cognition is bi-directional, which means that the physical humans and the virtual agent all have their cognitive ability. Therefore, attention is also considered by both physical humans and virtual agents.

2.1.2 Active Search

Considering the uncertainty of the states of physical environments, humans need to actively sense the surrounding environments to know whether there is a change caused by virtual agents. The same thing happens to virtual agents. The virtual environment is also in a changing state, which is caused by physical humans. Based on the above considerations, both the physical humans and the virtual agents should implement their cognition by active search [3].

Active search requires the agent (physical human or virtual agent) actively search the surrounding environment to find valuable objects or events, and then learn knowledge from the observation. A passive search cannot meet the demand because changes may be caused at any time and any place due to the physical-virtual mapping within a symmetrical reality system.

2.2 Perception

Humans perceive the world by vision, audition, haptics, *etc.*, and human perception [2] has been studied for many years. However, we will talk about the perception within symmetrical reality. The perception in symmetrical reality refers to two perceptual centers,

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i.e., the physical human and the virtual agent. Symmetrical reality emphasizes the bi-directional interaction between the physical human and the virtual agent.

From the physical human's perspective, the virtual world is something augmented to the physical world. When the human can only perceive the physical world, it is called "physical reality". When the human can perceive both the physical world and the virtual world, it is called "augmented reality". When the human can only perceive the virtual world, it is called "virtual reality".

From the virtual agent's perspective, the physical world is something augmented to the virtual world. When the virtual agent can only perceive the virtual world, it is treated as a mirror of the physical reality, and we can just name it as "inverse physical reality". When the virtual agent can perceive both the virtual world and the physical world, it is called "inverse augmented reality". When the virtual agent can only perceive the physical world, it is called "inverse virtual reality". The perceptual form for virtual agents is the inverse direction of that for physical humans, so the inverse perception is the basic feature of a virtual agent-centered system.

Based on the above analysis, symmetrical reality can be understood as a kind of double-center perception systems supporting bi-directional interaction between physical and virtual worlds.

2.3 Pattern Recognition

Pattern recognition [5] is the crucial part that assists people to adapt to the world. In the complete cognition pipeline, people need to recognize the perceived information and apply it in new situations. This kind of information includes vision, audition, and haptics. Within the area of symmetrical reality, both the physical human and the virtual agent has the ability of pattern recognition, so the recognition issue is a little different from the traditional recognition issue. Note that this paper is aimed to introduce the basic concepts of pattern recognition in symmetrical reality, so we will not mention the specific technical algorithms.

We take visual recognition as an example to discuss the pattern recognition problem. If there is a cup of water on the top of a table in the physical world, the human can recognize the cup as a physical object according to our general understandings. However, symmetrical reality creates a new possibility for this situation. The cup may have its corresponding virtual embodiment in the virtual world, so that the virtual agent can also see the virtual cup as a virtual object. As for the recognition process, the physical human may probabilistically recognize the cup as an independent cup or a mapping cup, while the virtual agent also has the probabilistic knowledge. The method to determine whether this cup is independent or not is to actively interact with it. If the physical cup and the virtual cup change simultaneously in any situation, it means the two cups are related to each other and the behaviors will be affected by the symmetrical reality system; otherwise, they are independent and the behaviors will not be affected.

2.4 Communication

The communication between the humans and the virtual agent is an important way to make the symmetrical reality system run better. Some researchers do the explanations of robot behaviors to enhance the interaction between the human and the robots [4]. Of course, robots can be treated as the physical embodiment of virtual agents. The communication between the physical human and the virtual agent will promote the cognition process by knowledge exchange, especially for those collaboration tasks.

Communication is the knowledge-based interaction between different people or virtual agents. In a symmetrical reality system, the physical human and the virtual agent can help each other with certain tasks based on the advantages of each world. Noticing that the virtual world and the physical world are not the same as each other, there are correspondences between the two worlds and the private setting of each. For example, if we want to learn to fold

clothes, we could easily get virtual guiding information from the virtual world and successfully finish the task. If the virtual agent wants to fold clothes but does not know how to do it, physical humans can share the knowledge to assist the virtual agent. This communication process can be guaranteed by the symmetrical reality framework.

3 APPLICATION OF SYMMETRICAL COGNITION

The cognition within the symmetrical reality system is called symmetrical cognition. Symmetrical cognition could be used in many fields, such as commonsense inference and communicative learning.

3.1 Commonsense Inference

Learning commonsense is the basic ability to live in the world. Symmetrical cognition can play a role in commonsense inference by encoding knowledge in both physical and virtual worlds. The virtual world can be set as a fully accessible environment, so that when humans are experiencing some events, they can get the ground truth information from the virtual world. This setting can help the physical human learn the commonsense more efficiently due to the full ground truth. On the contrary, if the virtual agents (or robots) want to learn to do some tasks, they can get help from the physical humans in the physical world.

3.2 Communicative Learning

Communicative learning requires at least two minds to communicate with each other. This is in accordance with the framework of symmetrical reality. Within a symmetrical reality system, the virtual agent and the physical human can communicate with each other by penetrating the boundary between the virtual world and the physical world. The information could be exchanged in different ways, including visual images, audio information, force, *etc.* Therefore, the symmetrical reality framework is useful to promote the research and application of communicative learning.

4 CONCLUSION

We briefly discuss the cognitive issues within the framework of symmetrical reality. There are four parts that may affect cognition, *i.e.*, attention, perception, recognition, and communication. For each part, we have presented the specific points caused by symmetrical cognition. In future research, experiments will be conducted to demonstrate how symmetrical reality can affect the cognition of physical humans and virtual agents in detail.

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